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AN APPLICATION OF THE COX PROPORTIONAL HAZARDS MODEL TO BANK FAILURE

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The purpose of this study is to present the Cox proportional hazards model and to apply this model to the prediction of bank failures. The Cox model, which has been used extensively in biomedical applications, has not been previously employed in the finance literature. The principal advantage of the Cox model over other classification techniques is that it models the expected time to failure. Results of the study indicate that total classification accuracy of the Cox model is similar to that of discriminant analysis, although the Cox model produces somewhat lower type I errors. In a comparison of actual and predicted times to failure, the Cox model tends to identify bankruptcies prior to the actual failure date.

1. Introduction

The recent increase in the failure rate of commercial banks in the United States has refocused attention on efforts to identify 'problem' banks and to predict failures with sufficient lead time for regulators and management to institute remedial action at these banks. In the mid-1970s, bank regulatory agencies began using computerized statistical models as off-site 'early warning systems' to assist in the identification of likely candidates for failure. These models help the agencies identify particular banks for on-site examinations and isolate particular problem areas.

One common multivariate approach used to identify problem banks has been multiple discriminant analysis (MDA). Although MDA has been shown to predict bankruptcy with a relatively high level of accuracy, this technique has several shortcomings, which have been discussed by Altman et al. (1981), Eisenbeis (1977), Joy and Tollefson (1975) and others. One neglected drawback in using MDA to predict bank failures is that the output from discriminant analysis is limited to the posterior probability that a particular bank will fail, given its financial characteristics. The expected time to failure is not given explicitly. For example, if a discriminant model is based on data

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taken one year prior to failure, the output from the model is the posterior probability that a particular bank will fail *within one year*. Examinations scheduled from these probabilities may not be in time to forestall a failure that would actually occur in much less than one year. Furthermore, mixed time models cannot be developed with MDA. In constructing the model, observations one year prior to failure cannot be mixed with observations, even of different banks, made two years prior to failure. Similarly, models developed from data one year prior to failure cannot be used to predict failure two years in advance.

The omission of time to failure lessens the usefulness of MDA and its usual alternatives (logit, probit, and regression analyses) to regulatory agencies. In addition, the parametric nature of these procedures requires certain assumptions that are often difficult to meet in most empirical applications, thereby possibly reducing their effectiveness.

This paper focuses on a procedure, the Cox (1972) proportional hazards model, which addresses both of these issues. Applied extensively in biomedical studies, especially with heart transplant and cancer data [e.g. Crowley and Hu (1977)], this model is particularly useful in the context of predicting bank failure because of its lack of underlying assumptions and its ability to provide an estimate of the probable time to failure. The usefulness of the model as part of an early warning system for commercial banks is examined in this paper with respect to both accuracy in predicting time until failure and classification error rates. Thus, the major purpose of the paper is to present a new classification technique, the Cox model, and to highlight its advantages, namely (1) its lack of underlying assumptions and (2) the ability of the model to incorporate explicitly the time to failure into the modeling effort.[1] Multiple discriminant analysis provides the benchmark for evaluation of the classification ability of the Cox model.

Section 2 of the paper presents a brief description of the Cox model. In section 3 the sample and variables used in the empirical analysis are presented. Fitting the Cox model is described in section 4, and its classification ability is compared to that of MDA in section 5. The ability of the fitted model to estimate likely time to failure is examined in section 6. Section 7 presents a summary and conclusions.

2. The Cox proportional hazards model

Let T indicate the time to failure of an individual bank. Then $S(t) = \Pr[T > t]$, the probability that a bank will live longer than t time units, is

[1]A nonparametric classification technique, recursive partitioning, has been proposed by Frydman et al. (1985) for prediction of bankrupt firms. Like its parametric alternatives, this procedure does not explicitly consider the likely time to failure. Additionally, West (1985) has proposed a new approach to early warning systems for commercial banks using factor analysis and logit estimation.

called the survivor function. The distribution function of time to failure is given by $F(t)=1-S(t)$ and the density function is $f(t)=-S'(t)$. While the distribution of time to failure could be described in terms of $F(t)$ or $f(t)$, it is more commonly characterized by the hazard function:

$$h(t)=\lim_{\Delta t\to 0}\frac{P[t<T<t+\Delta t|T>t]}{\Delta t}=\frac{-S'(t)}{S(t)}.$$

The hazard, $h(t)$, is the probability of failure in the next instant, given that the bank was alive at time t. In actuarial statistics, $h(t)$ is called the force of mortality and in economics its reciprocal is called Mill's ratio.

The present work is concerned with modeling the hazard function for banks in the United States. There are several technical advantages in estimating $h(t)$ rather than $F(t)$ or $f(t)$; these are discussed in Cox and Oakes (1984, p. 16) and Lawless (1982, p. 11). Of course, once an estimate of $h(t)$ has been obtained, estimates of $f(t)$ and $F(t)$ are readily available from

$$F(t)=1-\exp\left[-\int_0^t h(u)\,\mathrm{d}u\right]$$

and $f(t)=F'(t)$.

It will be assumed in this paper that the appropriate form for the hazard function of banks is given by the proportional hazards model (PHM). Let $h(t|z)$ denote the hazard function at time t for a bank with explanatory variable vector z. Under the PHM, $h(t|z)=\psi(z)h_0(t)$, where $\psi(z)$ is some function of z such that $\psi(0)=1$ and $h_0(t)$ is the underlying hazard, i.e. the hazard function of a bank with $z=0$. If the explanatory variables have been centralized, so that a bank with $z=0$ has values equal to population means, then $h_0(t)$ can be thought of as the hazard function for an 'average' bank in the population. Thus, the primary assumption underlying the PHM is that the effect of the explanatory variables is to multiply the hazard of an average bank, $h_0(t)$, by some function, $\psi(z)$, of the deviations of the explanatory variables from their mean values. There is empirical evidence to support this assumption in many different fields of scientific endeavor [Cox and Oakes (1984, p. 71)]. In addition, there are technical reasons for using the PHM: (1) it can easily accommodate incomplete (censored) failure data, in which the time to failure is unknown for certain banks and (2) statistical estimation and inference problems for the model have a rather simple solution even when $h_0(t)$ is arbitrary. For further discussion of the advantages of the PHM, see Cox and Oakes (1984, pp. 70–71) and Kalbfleisch and Prentice (1980, pp. 32–33).

The most important special case of the PHM is the model proposed by Cox (1972) in which $\psi(z)=\exp(\beta' z)$, where β is a vector of regression coefficients. The hazard function is then given by

$$h(t|z)=\exp(\beta' z)h_0(t). \tag{1}$$

The model for the time to failure with hazard function given by (1) will be referred to as the Cox model throughout the remainder of this paper. The primary reason for choosing an exponential function for $\psi(z)$ is that such a choice greatly simplifies the estimation of the regression coefficients β [Cox and Oakes (1984)].

The Cox model cannot truly be called nonparametric since it does depend on the regression parameter vector β. However, the baseline hazard function $h_0(t)$ is assumed to be arbitrary and no distributional assumptions are required to carry out the estimation of β or $h_0(t)$. Thus, the Cox model is often referred to as semiparametric; $h_0(t)$ is its nonparametric part and $\exp(\beta' z)$ is its parametric part.

In terms of the present study, the vector z corresponds to some set of accounting ratios, and t corresponds to the survival time of a bank in the study, as defined in the following section. The survivor function to be estimated here is

$$S(t|z)=S_0(t)^{\exp(\beta' z)},$$

where

$$S_0(t)=\exp\left[-\int_0^t h_0(u)\,\mathrm{d}u\right]$$

is the survivor function corresponding to the baseline hazard function $h_0(t)$.

As in discriminant analysis, fitting the Cox model[2] requires identifying a sample of (in this case) failed banks and a control sample of non-failed banks. Observations over a vector of financial variables are made for both samples k periods prior to failure, and the Cox model is fit to these observations. The resulting survivor function provides an estimate of the probability that a particular financial institution possessing certain financial characteristics will survive for t periods into the future, where $t \leqq k$.

[2]The procedure *PHGLM* of the Statistical Analysis System (SAS) is used in fitting the Cox model. Computational details on the algorithms used in this procedure are contained in the SUGI Supplemental Library User's Guide [SAS Institute Inc. (1983, pp. 268–271)].

3. The sample and variables

3.1. Sample selection

A sample of failed commercial banks is selected from listings appearing in the 1979 through 1983 annual reports of the Federal Deposit Insurance Corporation. Additionally, failures through June 1984 are identified from an interim publication of the FDIC. Thus, the sample includes all failed banks in the United States from January 1979 through June 1984 for which complete data were available. The date of failure is included in these lists and represents the official date of closure, either when the failed bank was assumed by another bank or when there was a payoff of depositors.

The development of a statistical model of bank failure requires (1) identification of a sample of peer banks to serve as a basis of comparison with the failed banks, and (2) accounting data for the banks in question for several years prior to failure. The most readily accessible source of financial statement data on individual commercial banks is the FDIC/Federal Reserve System subscription data tapes. Each failed bank in the sample is required to have annual data on the FDIC tapes for at least two years prior to the year of failure.

To provide a basis for comparison, an attempt is made to match each failed bank with at least one nonfailed bank[3] on the basis of five criteria:

(a) geographic location;
(b) charter status (state or national);
(c) size;
(d) holding company affiliation; and
(e) age.

For failed banks in Standard Metropolitan Statistical Areas, matches are selected from the same SMSA. If the failed bank is not located in a SMSA, the match is selected from the same geographical (non-SMSA) area. Total assets is used as a proxy for size, and the total assets of a bank are not allowed to exceed twice those of its matching bank.

The final sample consists of 130 banks that failed between January 1978 and June 1984 and 334 matching nonfailed banks. Selected characteristics of the final sample are given in table 1. Failed banks typically are small, state-chartered institutions, although large banks are not immune to failure.

3.2. Variable selection

Off-site analysis of the financial condition of commercial banks is restricted

[3]Adequate matches could not be found for 18 failed banks; the largest number of matched banks for any one failure is ten. Although matching is an inferior sampling technique to random sampling, the matched sampling technique is used in this paper because a random sample of banks would contain too few failed banks to apply either the Cox model or MDA.

Table 1
Characteristics of the sample one year prior to failure.

	Number of banks	Mean net income	Mean total assets	Mean total deposits	Number with deposits (in $ million) ≦$20	≧$200
Failed	130	$−1.161	$46.984	$42.090	68	5
Nonfailed	334	$ 0.299	$44.191	$38.752	152	7

All dollar figures are in millions.

generally to their financial statements. Market data such as stock prices are available only for a few hundred of the largest institutions, and even then the stock may be of the larger holding company, not the bank itself.[4] To be applicable to the population of banks, early warning systems, therefore, rely on accounting data. An examination by the FDIC, the Federal Reserve Board, or the Office of the Comptroller of the Currency focuses on five general areas: capital, asset quality, management, earnings quality and amount, and liquidity. These areas are frequently identified by the acronym CAMEL, from the first letter of each, and were adopted in 1978 as the foundation for a uniform approach to examination. There is little agreement among the three agencies, however, as to which financial ratios are best for screening the data off-site. Nor is there a consensus in the academic literature on which ratios are the best predictors of failure.[5] Compounding the problem are the high correlations among accounting ratios.

Twenty-one ratios are employed in this study as potential predictors of failure. These ratios are listed and defined in table 2. Ratios such as these have appeared in previous studies and, with few exceptions, have been identified as good indicators of failure. Reviews of this literature are found in Sinkey (1977a, 1977b, 1979), Santomero (1983), and Bovenzi et al. (1983). These 21 ratios are chosen so that each of the CAMEL areas is represented. Because these areas are not clearly defined, a large number of ratios covering several facets of each are included. The dimensions labeled 'loan composition' and 'efficiency' are assumed to measure managerial ability to the extent that they reflect explicit managerial decisions.

Although the Cox model does not assume multivariate normality, the benchmark procedure of MDA does. As a first step in verifying the assumption of multivariate normality of the 21 ratios, a univariate test for normality was performed on each ratio using a modified Kolmogorov–Smirnov test

[4]Altman et al. (1981) and Shick and Sherman (1980) have shown that information derived from bank stock prices is useful in constructing an early warning system. Since most of the banks in this study are small, their stock prices are generally not available.

[5]Sinkey (1977b) indicates that, in bank failure prediction, capturing certain dimensionalities is much more important than the form that the individual variables take.

on the data for the nonfailed firms for both one year prior ($n=334$) and two years prior ($n=301$) to failure. For each of the non-normally distributed ratios, a log transformation was applied, achieving normality for one ratio and improving the normality test results for several others,[6] as noted in table 2. Since a majority of these ratios are non-normally distributed, even after transformation, the hypothesis of multivariate normality is rejected for the data in this study.

For each failed bank in the study and each nonfailed bank matched with it, values are obtained for the 21 ratios as of 31 December of the year prior to failure of the failed bank. These values constitute what is referred to as the one year prior data set. Similarly, the two years prior data set contains the

Table 2

Variable definitions, dimensions, abbreviations, and transformations.

Dimension	Ratio number	Abbreviation	Variable definition
Capital (leverage)	11	*TCTA*	log total capital/total assets
	12	*TLTC*	log total loans/total capital
	3	*FSTA*	log fed. funds sold + securities purchased/total assets
Asset (loan) quality	15	*LRTL*	net loan recoveries/total loans
	9	*PLOE*	log provision for loan losses/total operating expense
	10	*CONI*	log gross loan charge-offs/net income + provisions of loan losses[a]
Loan composition	6	*CLTL*	log commercial and industrial loans/total loans
	7	*RLTL*	real estate loans/total loans[b]
Efficiency (pricing)	16	*LRNL*	loan revenue/net loans
	14	*OITA*	total operating income/total assets
	17	*IDTD*	interest on deposits/total time and savings deposits[c]
Earnings	20	*TEBT*	income taxes/earnings before taxes and security transactions
	13	*OEOI*	log total operating expense/total operating income
	18	*NITA*	net income/total assets
	19	*NITC*	net income/total capital
	21	*NIOI*	net income/gross operating income
Liquidity	8	*LODE*	total loans/total deposits[b]
	5	*LOTA*	total loans/total assets[c]
	1	*CATA*	log cash and U.S. securities/total assets[b]
	2	*MSTA*	log municipal securities/total assets
	4	*FPTA*	log fed. funds purchased + securities sold/total assets[d]

[a]Transformed for one year prior only.
[b]Normally distributed.
[c]Normally distributed for one year prior only.
[d]Transformed for two years prior only.

[6]Other transformations were examined, and the log transformation provided the most improvement in the test statistic.

values of the same ratios, as calculated on 31 December two years prior to failure.[7]

3.3. Sample considerations

One of the reasons for the popularity of the Cox model in medical applications is that 'censored' as well as complete lifetime data can be used in building the model. It is frequently the case in studying a time to failure distribution that the sample consists not only of items whose time until failure is known, but also items for which only a lower bound for the time until failure is available due to some aspect of the design of the study. Such an item is said to have a censored survival time. Censoring usually applies to those observations that have not failed by the end of the study or which are withdrawn from the study for some reason other than failure. In the present work, for example, the survival times of all nonfailed banks in the sample are censored, since all that is known about these banks in terms of their lifetime is that they survived until 30 June 1984.

The survival time for a failed bank is defined to be the time (in months) from 31 December of the year that the financial ratio data were reported to the FDIC (either one or two years prior to failure) until the date of failure. For a nonfailed bank, the censored survival time is defined to be the time (in months) from the above reporting date until 31 December of the year of failure of its matched failed bank.[8]

Two important considerations in designing a sample to be used with the Cox model are (1) the presence of ties among the observed survival times and (2) the time dependence of the variables z. With regard to (1), the Breslow (1974) correction for ties is used in fitting the Cox model to the data in this study since all the nonfailed banks have the same censored survival time of one or two years, as defined above. With regard to (2), the assumption is made in this study that the values of the independent variables (accounting ratios) for a particular bank remain constant over the time interval that the bank is in the study. For example, for the data collected one

[7]On average, the 'one year prior' data are observed 6 months prior to failure and the 'two year prior' data are observed 18 months prior to failure (assuming failures are evenly distributed throughout the year). Since the objectives of this paper are expositional, the issue of what time periods prior to failure should be used in developing the model is not addressed. In a practical application, two years prior is probably more desirable than one year due to the time lag associated with reporting accounting data.

[8]A listing of the survival times and ratio values for all the failed and nonfailed banks in the sample is available from the authors. The censored survival times are based on the failure of the paired bank because of the assumption underlying the Cox model that accounting ratios for a particular bank remain constant over the time interval that the bank is in the study. A Cox model is also fit using a longer censored survival time (defined to be the time from the reporting date until 30 June 1984), and the classification results are identical to those presented in this paper.

year prior to failure, this assumption implies that the values of the 21 ratios do not change from 31 December of the year that the data were reported for a particular bank until either the bank failed or was censored. Although constancy of the ratios over a 12-month period may be questioned, the goodness of fit and prediction results described in the following sections indicate that even if the assumption of constant ratios is not valid for these data, the Cox model based on this assumption still provides useful results.

4. Fitting the Cox model

To determine the 'best' subset of the 21 ratios, a stepwise procedure combining forward and backward elimination is applied to both the one year and two years prior data for the banks that failed in 1979–1983, together with their matches [e.g. Bartolucci and Fraser (1977)]. For the one year prior data, the following ratios are retained: *CLTL*, *LODE*, *TCTA*, *OEOI*, and *NITA*. The ratios *NITA* and *OEOI* have a strong negative correlation, −0.836, causing the coefficient of *NITA* to have an unexpected sign (positive instead of negative) in the Cox regression equation. Therefore, the ratio *NITA* is dropped from the model.[9]

For the two years prior data, the stepwise procedure yielded *MSTA*, *LOTA*, *CLTL*, *OEOI*, *NITC*, *TCTA* and *TEBT*. Two of the ratios are highly correlated: *TEBT* and *OEOI* (−0.904). This collinearity yields an insignificant result for the test of the coefficient of *TEBT* ($p=0.7230$). Hence, this ratio is eliminated. Cox models are then fit to the four-ratio set for the one year prior data and the six-ratio set for the two years prior data, using the banks that failed in 1979–1983, together with their matches.[10]

Table 3 contains a summary of the estimation results. For each variable in these models, the table gives the estimate of the coefficient, the probability level for the test that the population coefficient is equal to zero, and the partial R-value, which is a measure of the contribution of the individual variable to the predictive power of the overall model. A positive coefficient in the Cox model indicates that an increase (decrease) in the variable is associated with a decrease (increase) in the survival probability. The signs are all as expected with the exception of that for *NITC* in the two year prior results. This unexpected sign is likely the effect of undetected multicollinearity among the variables in the model. However, *NITC* is retained in the Cox model since it entered before *OEOI* in the stepwise selection procedure.

[9]While collinearity may improve the predictive ability of MDA models in certain instances, it should be avoided whenever possible when using the Cox model, as shown by Hearne et al. (1983).

[10]The choice of these ratios is consistent with the results of a principal components analysis of the two data sets. This is not unexpected since the ratios are selected for this paper primarily from the results of earlier studies, some of which employ similar variable reduction techniques.

Table 3
Summary of results for Cox models.

Variable number	Abbreviation	Coefficient	Probability level	Partial R
A. One year prior data				
6	*CLTL*	0.390	$\leqq 0.0001$	0.116
8	*LODE*	0.747	$\leqq 0.0001$	0.172
11	*TCTA*	−1.006	$\leqq 0.0001$	−0.251
13	*OEOI*	0.744	$\leqq 0.0001$	0.232
B. Two year prior data				
2	*MSTA*	−0.381	0.0165	−0.065
5	*LOTA*	0.816	$\leqq 0.0001$	0.156
6	*CLTL*	0.446	$\leqq 0.0001$	0.118
11	*TCTA*	−0.648	$\leqq 0.0001$	−0.148
13	*OEOI*	1.013	$\leqq 0.0001$	0.212
19	*NITC*	1.929	0.0007	0.103

Taken as a whole, the results in table 3 indicate that all of the coefficients in these models are highly significant and that the individual variables have a fairly high degree of predictive ability.

Once the Cox methodology has been applied to an appropriate set of ratios, the next step is to test for goodness of fit of the proposed model. A chi-square likelihood ratio test of the significance of the overall model [Schoenfeld (1980)] indicates that both Cox models are highly significant ($p \leqq 0.0001$). A descriptive statistic that is useful in assessing the goodness of fit and predictive ability of a Cox model is the R statistic, which is a generalized version of the r^2 statistic in multiple regression analysis and is based on the Akaike information criterion [e.g. Atkinson (1980)]. The R-values for the Cox models are 0.449 for the one year prior data and 0.350 for the two years prior data, both of which are indicative of a reasonably good fit.

There are also two graphical techniques for testing the goodness of fit of the Cox model. One involves the generalized residuals of the fitted model, which are defined by $e_i = -\ln \hat{S}(t_i|z_i)$, $1 \leqq i \leqq n$, where n is the number of failed banks and $\hat{S}(t_i|z_i)$ is the estimated survival probability for bank i with vector z_i that failed at time t_i. These residuals are ordered from smallest to largest and plotted against the estimated cumulative hazard function $\hat{H}(e_i)$, which is given by the formula on p. 685 of Dixon (1981). If the Cox model is appropriate, the resulting plot should be a straight line of slope one [e.g. Kay (1977)]. The plots for the four variable model for the one year prior data and for the six variable model for the two years prior data are given in figs. 1(a) and 1(b), respectively. Both plots indicate an adequate degree of fit for the corresponding models.

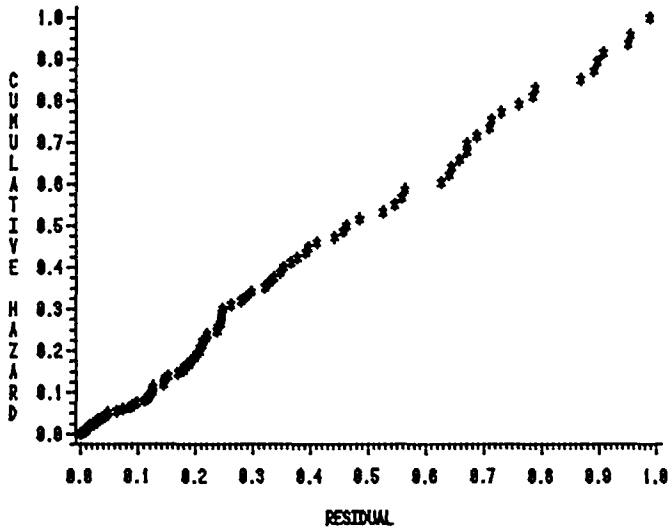

Fig. 1. (a) Goodness of fit plot – one year prior.

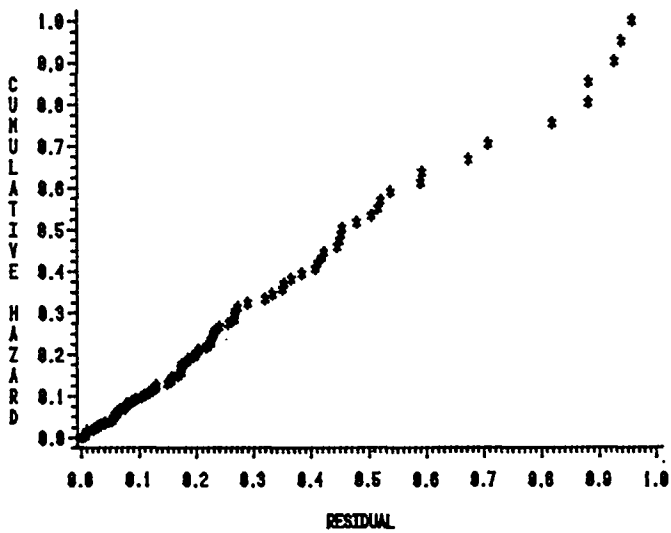

Fig. 1. (b) Goodness of fit plot – two years prior.

The other graphical technique for checking the goodness of fit of the Cox model involves a verification of the assumption of proportional hazards. This is accomplished by stratifying each covariate into a number of levels and then fitting a Cox model separately within each stratum. A separate plot is then constructed of $\log[-\log \hat{S}_0(t_i)]$ versus t_i in each stratum. If the resulting graphs are approximately parallel across strata, then the proportionality assumption is assumed to be valid [e.g. Lawless (1982, p. 279)]. Such plots are constructed for each variable in the four-variable model for the one year prior data and the six-variable model for the two years prior data and the assumption seems to be valid in each case. Fig. 2 contains such a plot for the variable *TCTA* for the one year prior data.

5. Comparison of Cox model and MDA

To be useful as part of an early warning system, the Cox model must be able to discriminate between sound banks and those likely to fail. The classification ability of the Cox model is compared in this section to that of MDA. The MDA and Cox models are constructed using only the data for banks that failed during 1979–1983, together with their matches. These models are then evaluated in terms of their accuracy of prediction of failure for the banks that failed from 1 January 1984 through 30 June 1984, together with their matches. Both types of models are constructed for both data sets.

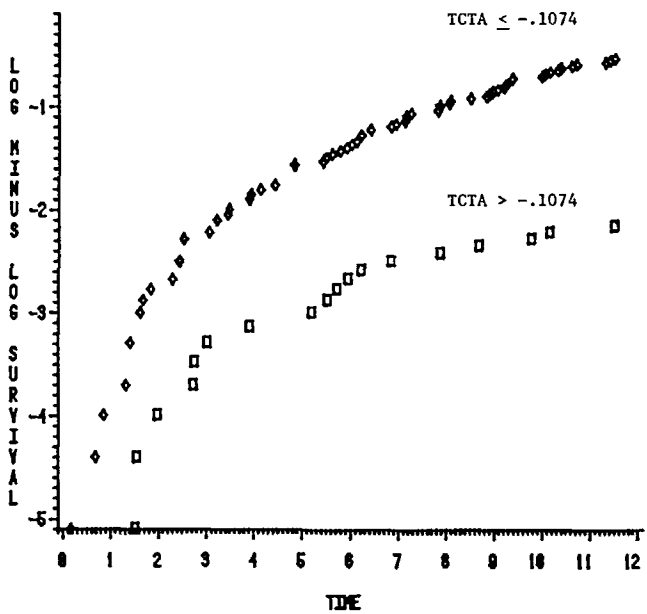

Fig. 2. Log (−log) plot for one year prior.

5.1. Multiple discriminant analysis

To determine those subsets of the 21 ratios that possess the best predictive ability in an MDA model, both forward and backward elimination techniques are applied to the entire set of 21 ratios for both sets of data. The following four ratios are retained for the one year prior data: *CLTL*, *LODE*, *TCTA* and *OEOI*. These are the same ratios retained in the Cox model for this data set.

For the two years prior data, the ratios *NITC*, *OEOI*, *LOTA*, *LODE*, *CLTL* and *MSTA* are retained in the MDA model. Five of these ratios were selected for the Cox model, with *TCTA* in the Cox model being replaced by *LODE* in the MDA model.

Once a subset of ratios has been chosen, it becomes necessary to determine whether linear or quadratic MDA should be used. Theoretically, quadratic analysis should always be used when the population covariance matrices are unequal [Johnson and Wichern (1982, p. 484)]. For comparison purposes, however, both linear and quadratic analyses are performed in this study.[11]

Another aspect of using MDA is the specification of the prior probabilities of group membership. These probabilities should be set equal to a reasonable estimate of the probabilities of group membership in the population, assuming that the sample is random [Eisenbeis (1977, p. 890)]. The data in this study, however, do not constitute a random sample since they are obtained by matching every bank that failed between 1 January 1979 and 30 June 1984 with one or more comparable nonfailed banks. Under these circumstances, the sample proportions (0.29 for failed banks, 0.71 for nonfailed) are used as the prior probabilities. This choice of proportional priors prevents any generalization about the accuracy of the classification model to the population [Eisenbeis (1977, p. 890)]; however, results based on such priors can still be used in comparing various classification schemes. Proportional priors are used throughout this study, since they are appropriate for a matched sample [Eisenbeis (1977, p. 890)].

5.2. Classification results

While the Cox model has been used extensively in modeling the relationship between survival and independent variables in biomedical applications, no studies could be found in which the model had been used for prediction

[11] Box's modified M criterion [Morrison (1976, p. 252)] is used to test the assumption of equal covariance matrices between the two groups and this hypothesis is rejected ($p<0.0001$) for both data sets. However, these results may be misleading since financial ratio data is typically nonnormally distributed, as indicated by the tests on the data in the present study and the results of other authors [e.g. Deakin (1976)]. Box's criterion is not very robust in the presence of such data [Morrison (1976, p. 253)] and therefore may lead to erroneous conclusions about the equality of the covariance matrices. Therefore, both linear and quadratic MDA models are considered in this study.

or classification. The classification procedure proposed here is to calculate the probability that a particular bank will survive longer than t months, given the values of the financial ratios for the bank, where $t=12$ for the model based on the one year prior data and $t=24$ for the model based on the two years prior data. If this probability is less than some cutoff value, C, then the bank is classified as a likely failure. Otherwise, the firm is classified as 'healthy', or unlikely to fail. Determination of the value of C is critical to the performance of such a classification rule. Since the output from the Cox model, $S(t|z)=P[T>t|z]$, is the probability that a bank will not fail during the next t time units, the choice of C is similar to a problem encountered in logit analysis, in which the output is $P[Y=0]$, where $Y=1$ if the bank fails and 0 otherwise. For logit analysis, Martin (1977) recommends a value of C equal to the sample proportion of nonfailed banks. Therefore, this value for the 1979–1983 sample is used in the classification scheme for the Cox models in this study as well. This is equivalent to specifying proportional priors and equal misclassification costs for the MDA model.

The classification results for the MDA models with proportional priors are shown in table 4. The results for the one year prior data indicate a total classification accuracy of approximately 89 percent using linear rules and 86 percent using quadratic classification rules. Using two years prior data, the results degrade to an overall correct classification accuracy of approximately 73 and 65 percent for linear and quadratic rules, respectively. These results are comparable to those from previous studies using MDA [Sinkey (1979) and Altman et al. (1981)].[12]

The classification results for the Cox models using sample proportions as cutoff points are also given in table 4. As in previous failure prediction studies, a type I error is defined to be the misclassification of a failed bank as nonfailed and a type II error is defined to be the misclassification of a nonfailed bank as failed. Examination of the results in table 4 indicates that the numerical classification results from the Cox model compare favorably with those from MDA, especially for the model using two years prior data. Statistically, however, neither model dominates. For example, in the two year prior data, the Cox model has a significantly lower type I error rate while linear discriminant analysis possesses a lower type II error rate. No significant difference exists in total classification accuracy.[13,14]

[12]Lachenbruch, Sneeringer and Revo (1973) showed via simulation that use of MDA techniques on nonnormal data can result in distorted estimates of the misclassification rates. Thus, the true error rates for the MDA model may be quite different from those reported in table 4.

[13]Statistical tests for differences in error rates are based on the Cochran Q test. See Sokal and Rohlf (1981) for a discussion of this procedure. Results of these tests are available from the authors upon request.

[14]Changing the priors or incorporating costs of misclassification into the modeling effort could substantially alter the relative classification accuracies of the Cox and MDA models

Table 4
Classification results.

			Error rates[a]					
			Holdout			Original sample		
Data set	Variables	Model type	Type 1	Type 2	Total	Type 1	Type 2	Total
One year prior	6, 8, 11, 13	linear MDA	0.08	0.12	0.11	0.38	0.03	0.13
		quadratic MDA	0.13	0.14	0.14	0.33	0.05	0.13
		Cox	0.03	0.26	0.20	0.13	0.11	0.11
Two years prior	2, 5, 6, 8, 13, 19	linear MDA	0.39	0.22	0.27	0.38	0.08	0.16
		quadratic MDA	0.26	0.39	0.35	0.13	0.24	0.21
	2, 5, 6, 11, 13, 19	Cox	0.11	0.33	0.26	0.17	0.19	0.19

[a]Type 1 error refers to the misclassification of a failed bank as nonfailed; type 2 error refers to the misclassification of a nonfailed bank as failed.

A type I error is more important in an early warning system than either total classification accuracy or a type II error.[15] Sinkey (1979) suggests that banking authorities, given their goal of failure prediction, would be most concerned about the size of the type I error rate and would be willing to accept a higher type II error rate for a lower type I error rate.

6. Survivor profiles and time to failure

The evidence presented in table 4 indicates that, in general, the classification ability of the Cox model is comparable to that of discriminant analysis. The major contribution of the Cox model, however, lies in the additional information regarding the likely time to failure provided by the model. This additional information is captured in the estimated survivor function for a given bank with independent variable vector z. The Cox model yields an estimate of $S(t|z)=P[T>t|z]$, where T is the time to failure of the bank and t is a given number of time periods.[16] One can then plot the survival probability, $S(t|z)$, against t for a given covariate vector z, and the resulting curve can be used as a survival profile for the bank. Fig. 3 contains such a profile for the vector of average values for *CLTL*, *LODE*, *TCTA* and *OEOI* for both the failed and nonfailed banks in the one year prior holdout sample. The difference in the survival curves for the failed and nonfailed banks is dramatic. Fig. 4 contains the same plot for one selected failed bank (the West Coast Bank) and one of its matches (the Century Bank), both of Los Angeles, using the values of *CLTL*, *LODE*, *TCTA* and *OEOI* as calculated for these banks as of 31 December 1983. The West Coast Bank failed on 27 April 1984, and graphical interpolation using the survival profile for it indicates that the bank would have been predicted to fail by this date ($t=3.88$ months) with a probability of 0.975. In contrast, the only information of this sort that would have been available from an MDA model based on the same data is the posterior probability that the bank would fail by 31 December 1984 (0.9958 for the linear MDA model based on the same set of ratios). Thus, the

presented in table 4. In order to assess the effects of using other cutoff points, several different values were tried, and the relationships presented in table 4 were essentially unchanged. Details are available from the second author.

[15]Type II errors may be more important to a bank's management and owners if such errors produce unnecessary examinations. Identification of a sound bank as a failing bank could conceivably precipitate a run on deposits if publicly disclosed, and thereby create a failure. Recently Korobow and Stuhr (1985) proposed a new measure to evaluate early warning models in which one group is substantially smaller than the other. In such models the type I or II error rate may be high even though the total misclassification rate is low. Their measure corrects for this bias.

[16]In this study, the model fitted to the one year prior data can be used to estimate the probability that a bank will last longer than t months, where $0<t\leqq 12$. For the model based on the two years prior data, $12<t\leqq 24$, since extrapolation outside the range of the data used to construct the model is not allowed for the Cox model.

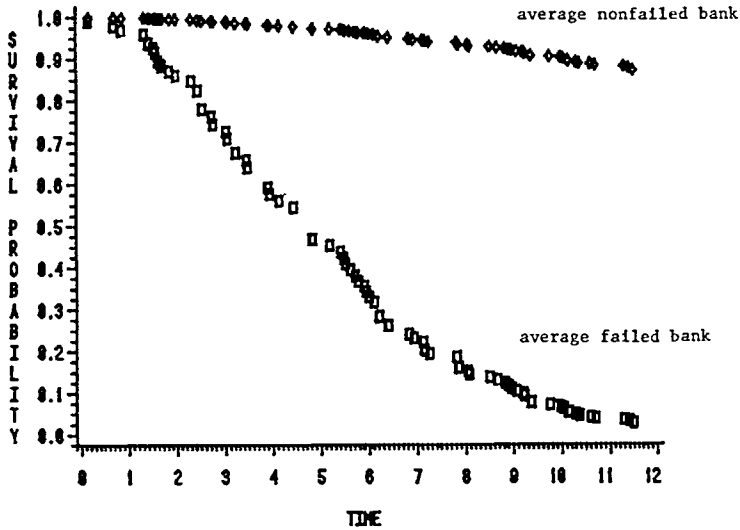

Fig. 3. Survival profiles – one year prior.

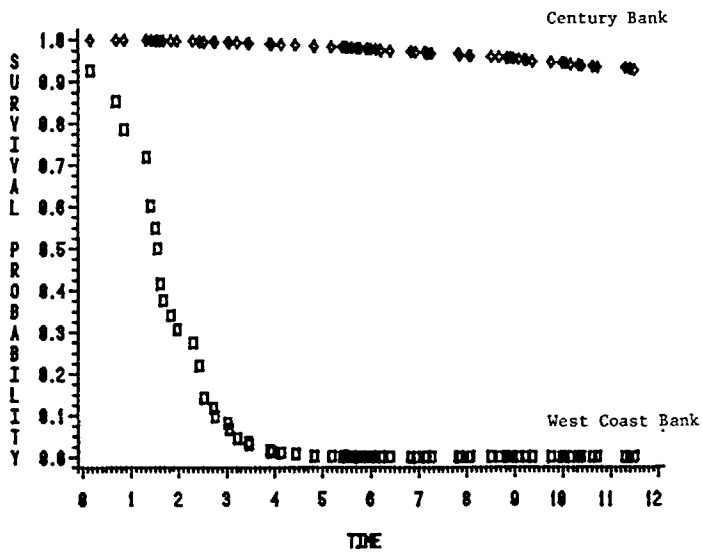

Fig. 4. Survival profiles – one year prior.

Cox model provides valuable information that could be helpful in making a timely identification of troubled banks, thereby facilitating the appropriate preventive action.

To illustrate further the uses of the survival profile, a profile based on the one year prior data is constructed for each of the 137 banks in the holdout sample. Estimated survival probabilities are calculated from this profile for intervals of one month starting with January 1984 and extending through December 1984. If the estimated survival probability for a given bank for a certain time period is less than the proportion of banks in the original sample that survived beyond that time period, then the bank is classified as a failure. Otherwise, the bank is classified as a nonfailure. For example, 99.082 percent of the banks in the original sample have survival times greater than one month. In the holdout sample, 38 banks have estimated one-month survival probabilities which are less than 0.99082 and are therefore predicted to fail by 31 January 1984. Of these 38 banks, two did fail by this date, 28 failed between 31 January and 30 June and eight survived beyond 30 June. The misclassification rates shown in table 5 indicate that the Cox model correctly identified both banks that failed in the first month of 1984 as likely failures in this month, i.e. the misclassification rate is zero. For the remaining 36 banks that eventually failed, however, 28 were also predicted to fail in the first month, and eight nonfailed banks were predicted to fail in the first month. The fact that the Cox model tends to identify banks as failures before their actual failure date is actually an advantage, since early identification of a troubled bank allows more time for taking the appropriate preventive action.

Another potential use of the survival profile derived from the Cox model is the prediction of the failure time for a particular bank. Such a prediction requires using a single summary measure to represent all of the information contained in the survival profile, and the median time to failure for each bank is one such measure. The rank order correlation between the median predicted time to failure from the Cox model and the actual time to failure is 0.58, which is significant at the 0.001 level. Instead of using median predicted time to failure, however, bank regulators may prefer to examine the entire survival profile for each bank and assess the probability of failure at any point in time in terms of the cost of on-site examinations versus the cost of misclassifications.

7. Summary and conclusions

This paper presents the Cox proportional hazards model and applies the model to the prediction of bank failures. The Cox model has several significant advantages over most previously employed classification models. First, since this methodology is essentially nonparametric, the Cox model is

Table 5

Comparison of actual failure times and predicted failure times – in months (holdout sample only).

Time period (months)	Misclassification rate[a]		
	Failures[b]	Eventual failures[c]	Nonfailures[d]
0–1	0/2 (0%)	28/36 (78%)	8/99 (8%)
0–2	0/4 (0%)	28/34 (82%)	8/99 (8%)
0–3	1/9 (11%)	25/29 (86%)	10/99 (10%)
0–4	1/17 (6%)	18/21 (86%)	11/99 (11%)
0–5	3/29 (10%)	8/9 (89%)	12/99 (12%)
0–6	3/38 (8%)	–	14/99 (14%)
0–7	2/38 (5%)	–	16/99 (16%)
0–8	2/38 (5%)	–	17/99 (17%)
0–9	1/38 (3%)	–	19/99 (19%)
0–10	1/38 (3%)	–	24/99 (24%)
0–11	1/38 (3%)	–	26/99 (26%)
0–12	1/38 (3%)	–	26/99 (26%)

[a]Numerator is number of banks misclassified in each time period. Figures in parentheses are percent misclassified.

[b]Includes all failed banks whose survival time is within the indicated limits.

[c]Includes all failed banks whose survival time is outside the indicated limits.

[d]Includes all nonfailed banks in the sample.

not vulnerable to the criticisms directed toward parametric techniques. Second, and perhaps of more importance to empiricists in finance and economics, the Cox model provides information regarding the expected time to failure not available from more traditional classification techniques. Thus, the greatest usefulness of the Cox model in any application may be in describing the relationship between survival time and various explanatory variables.

Results of this study indicate that the Cox model is useful in financial distress applications. Overall classification rates between the Cox model and discriminant analysis are similar, indicating that the researcher may not have to give up classification accuracy in order to get the additional information

provided by the Cox model on the likely time to failure. As part of an early warning system providing regulators with the likely time to failure, instead of just the probability of failure sometime within a large time interval, use of the Cox model would improve efficiency in the allocation of on-site examination resources.

Financial distress is, of course, only one application of the Cox model, and the model need not be applied in isolation from other classification techniques. For example, traditional classification techniques are currently being used in other areas such as the identification of likely merger candidates. Estimates of the expected time to merger would be valuable information to investors specializing in takeover targets.

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